

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

ZHIJIE FAN, JI LI, AND DMITRIY ZANIN

ABSTRACT.

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1. INTRODUCTION

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$, $N \geq 2$, and let G be the corresponding finite reflection group. Let $k : R \rightarrow \mathbb{R}_+$ be G -invariant multiplicity function. Let $R_{\text{Dunkl},j}$ be the j -th Dunkl-Riesz transform corresponding to R and k . Let w be the Dunkl weight corresponding to R and k . Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$.*

- (i) *If $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,*

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{R,k} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

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(ii) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_{R,k}^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalised reduced root system if $\|v\|^2 = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $\kappa : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

Define $d(x, y) = \min_{g \in G} |x - gy|$. Recall from [8, Chapter VII, proof of Theorem 2.12] that $d(x, y) = |x - y|$ iff x and y belong to the same chamber.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} \kappa(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{\kappa(v)},$$

where the implicit constant only depends on N, R and κ . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{\kappa(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} = T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \dots, N$.

2.2. Weyl chamber as simplicial cone.

Definition 2.1. A connected component of the set

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

Theorem 2.2. Let Γ be a chamber. There exists a linear isomorphism $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$ (with $N_1 + N_2 = N$).

2.3. System of dyadic cubes.

Notation 2.3. Let

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube in $\mathbb{D}_k$ is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube $Q \in \mathbb{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

2.4. Nearly weakly orthonormal sequences. Rochberg and Semmes [13] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [5, 10, 12, 3, 4, 11, 13, 15, 14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [10, Section 13] for a complete proof).

Definition 2.4 and Lemma 2.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

Definition 2.4. Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_Q\}_{Q \in \mathbb{D}}$ is called (r, c) -NWO if

- (i) $\text{supp}(e_Q) \subseteq cQ$ for every $Q \in \mathbb{D}$;
- (ii) $\|e_Q\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} \leq \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}}$ for every $Q \in \mathbb{D}$;

Lemma 2.5. Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $0 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} \leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty},$$

$$\begin{aligned} \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, \mathbf{w}(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, \mathbf{w}(x)dx)) \right) &\leq \\ &\leq C_{\mathbf{w}, r_1, c_1, r_2, c_2, p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left(\{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [10]. $\square$

Lemma 2.6. *Let $w \in A_2(\mathbb{R}^N)$. Let $1 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\left\| \left\{ \langle A e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{p,q}(\mathbb{D})} \leq c_{\mathbf{w}, r_1, c_1, r_2, c_2, p, q} \|A\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}.$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [10]. $\square$

3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

Theorem 3.1. *Let Γ be a chamber. For all $x \in \Gamma$ and for all $r > 0$, we have*

$$w(B(x, r) \cap \Gamma) \geq c_\kappa w(B(x, r)).$$

In particular, $(\Gamma, |\cdot|, w(x)dx)$ is a doubling metric measure space.

Notation 3.2. *Let $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$. Set*

$$w_\theta(x) = \prod_{A \subset \{1, \dots, N_1\}} \left(\sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

Lemma 3.3. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r)} w_\theta(y_1, \dots, y_{N_1}) dy \leq c_{N, \theta}^{(1)} r^N w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \subset \left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} &\int_{B(x, r)} w_\theta(y_1, \dots, y_{N_1}) dy \leq \\ &\leq \int_{\left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_\theta(y_1, \dots, y_{N_1}) dy = \\ &= \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_\theta(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ &\leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\ &= w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ &\leq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}. \end{aligned}$$

$\square$

Lemma 3.4. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x,r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq c_{N,\theta}^{(2)} r^N w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$x + [\frac{r}{2N}, \frac{r}{N}]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned} & \int_{B(x,r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\ & \geq \int_{x + [\frac{r}{2N}, \frac{r}{N}]^N} w_\theta(y_1, \dots, y_{N_1}) dy = \\ & = \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta(u) du \times (\frac{r}{2N})^{N_2} \geq \\ & \geq \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times (\frac{r}{2N})^{N_2} = \\ & = w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \geq \\ & \geq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}. \end{aligned}$$

□

Lemma 3.5. *There exists $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let A_v be the set of those $1 \leq k \leq N_1$ with $((\Phi^{-1})^*(v))_l$ being strictly positive. We, obviously, have

$$c_{\Phi,v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi,v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi,v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi,v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. □

Proof of Theorem 3.1. The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. □

4. ROCHBERG-SEMMES BASED ESTIMATE

Theorem 4.1. *Let $N \geq 2$, $1 \leq N_1 \leq N$ and $N_2 = N - N_1$. Let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Let $\mathbf{w} \in A_2(\Omega)$ be such that*

$$\sup_{x \in Q} \mathbf{w}(x) \leq c_w \frac{\mathbf{w}(Q)}{m(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Suppose that $K : \Omega \times \Omega \rightarrow \mathbb{C}$ satisfies

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{\mathbf{w}(B(x, |x - y|))}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator on $L_2(\Omega, \mathbf{w}(x)dx)$ with integral kernel K . Suppose $N > 2s$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}} \|f\|_{L_N(\Omega)}.$$

Lemma 4.2. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every x , there exists a unique $Q \in \mathbb{D}_j$ containing x . For this Q , we have $\ell(Q) = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x - y)) = \sum_{j \in \mathbb{Z}} \left(\sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x - y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right). \end{aligned}$$

If $x \in Q$ and $\psi(\frac{x-y}{\ell(Q)}) \neq 0$, then

$$x \in Q, \quad |x - y|_\infty \leq \ell(Q).$$

We obviously have $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|x - y|_\infty \leq \ell(Q)$. Thus, $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$. In other words, $y \in 3Q$. Thus,

$$\chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

This completes the proof. $\square$

Lemma 4.3. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y),$$

where

(i) $e_{k,l,Q}$ is supported in Q and $f_{k,l,Q}$ is supported in $3Q$ for every $Q \in \mathbb{D}$ with $Q \subset \Omega$;

(ii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)};$$

(iii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)};$$

Proof. For every $x, y \in \Omega$, we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \end{aligned}$$

For every $Q \in \mathbb{D}$ with $3Q \subset \Omega$, define the map

$$K_Q : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + a\ell(Q), c_Q + b\ell(Q))\psi(a-b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

If $Q \in \mathbb{D}$ is such that $3Q \not\subset \Omega$, then the definition of K_Q is the same, a also runs through $[-\frac{1}{2}, \frac{1}{2}]^N$, but the parallelepiped b runs through is slightly smaller. **to co-authors: if you wish, specify the domain of b also.** In what follows, we lighten the notations by pretending that b is always defined on $[-\frac{3}{2}, \frac{3}{2}]^N$.

It follows from the assumption on K that

$$\|K_Q\|_{C^{2N+2}([- \frac{3}{2}, \frac{3}{2}]^N \times [- \frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

Hence, there exists an extension $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k,l \in \mathbb{Z}^N} \text{coef}_{k,l,Q} e^{i\langle k,a \rangle} e^{i\langle l,b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k,l,Q}|^2 \leq \left(\frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k,l,Q}(x) = f(x)\chi_Q(x)e^{i\langle k, \frac{x-c_Q}{\ell(Q)} \rangle},$$

$$f_{k,l,Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,Q} \cdot \chi_{3Q}(y)e^{i\langle l, \frac{y-c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x)K(x, y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y).$$

Obviously,

$$\begin{aligned} \text{supp}(e_{k,l,Q}) &\subset Q, \quad \text{supp}(f_{k,l,Q}) \subset 3Q, \\ \|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} &= \|f\|_{L_r(Q, \mathbf{w}(x)dx)}, \\ \|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} &= (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}. \end{aligned}$$

□

Lemma 4.4. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K,\mathbf{w},r} \left\| \left\{ m(Q)^{\frac{1}{N}-\frac{1}{r}} \|f\|_{L_r(Q)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}.$$

Proof. Let $A_{k,l}$ be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.3, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))} \leq \\ &\leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))}. \end{aligned}$$

By Lemma 4.3, the sequences

$$\left\{ \mathbf{w}(Q)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,Q}}{\|f\|_{L_r(Q, \mathbf{w}(x)dx)}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}, \quad \left\{ \frac{c_{K,\theta}^{-1} \mathbf{w}(Q)^{\frac{1}{2}}}{\ell(Q)} f_{k,l,Q} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 2.4. Using Lemma 2.5, we conclude that

$$\begin{aligned} &\|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ &\leq c_{\theta,r,1,\infty,3,N} c_{K,\theta} \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}_+^N, \mathbf{w}(x)dx))} \leq \\ &\leq \frac{N}{N-1} c_{\theta,r,1,\infty,3,N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \end{aligned}$$

$$\times \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{N, \infty}.$$

Obviously,

$$\mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \leq \left(\frac{1}{\mathbf{w}(Q)} \cdot \sup_{x \in Q} \mathbf{w}(x) \right)^{\frac{1}{r}} \times \|f\|_{L_r(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof. $\square$

Proof of Theorem 4.1. Take $r \in (2, N)$. The assertion follows by combining Lemmas 4.4 and A.1. $\square$

5. PROOF OF SPECIAL CBIKEL ESTIMATE IN DUNKL SETTING

Lemma 5.1. *For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^N$, for every $t > 0$, we have*

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa, \alpha, \beta} t^{1 - \frac{|\alpha|_1 + |\beta|_1}{2}} d(x, y)^{-2} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2).$$

Proof. This is (slightly weakened form of) Theorem 3.1 in [2]. $\square$

Lemma 5.2. *Suppose $s + N > 2$. We have*

$$\int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq c_{\kappa, s} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s}.$$

Proof. By (2),

$$w(B(x, t^{\frac{1}{2}}))^{-1} \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \begin{cases} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} & \int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq \\ & \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_{d(x, y)^2}^\infty t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N dt = \\ & = c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp(-\frac{c_\kappa}{t}) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_1^\infty t^{-\frac{s+N}{2}} dt. \end{aligned}$$

$\square$

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

Lemma 5.3. *For every pair of multi-indices α, β , for every $1 \leq j \leq N$ with $N + |\alpha|_1 + |\beta|_1 > 3$, we have*

$$|(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{d(x, y)^{1 - |\alpha|_1 - |\beta|_1}}{w(B(x, d(x, y)))}, \quad x, y \in \mathbb{R}^N.$$

an issue here is that for $\alpha = \beta = 0$, this imposes the condition $N \geq 4$. it would be good to prove something like this under the assumption $N \geq 3$.

Proof. We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^\infty e^{-t\Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y) = \pi^{-\frac{1}{2}} \int_0^\infty h_t(x, y) \frac{dt}{t^{\frac{1}{2}}}.$$

Using Lemma 5.1, we estimate

$$\begin{aligned} |(\partial_x^\alpha \partial_y^\beta K_{R_{\text{Dunkl},j}})(x, y)| &\leq \\ &\leq \frac{c_{\kappa, \alpha, \beta}}{\pi^{\frac{1}{2}} d(x, y)^2} \int_0^\infty t^{-\frac{|\alpha|_1 + |\beta|_1 - 1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt. \end{aligned}$$

Applying Lemma 5.2, we conclude

$$\begin{aligned} |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| &\leq \\ &\leq \pi^{-\frac{1}{2}} c_{\kappa, \alpha, \beta} \times c_{\kappa, |\alpha|_1 + |\beta|_1 - 1} w(B(x, d(x, y)))^{-1} \times d(x, y)^{1 - |\alpha|_1 - |\beta|_1}. \end{aligned}$$

The assertion follows immediately. $\square$

Lemma 5.4. *For every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

Proof. We have

$$w(Q) \geq c_\theta \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if $x \in Q$, then $|x - c_Q|_\infty \leq \frac{1}{2} \ell(Q)$ and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$w(x) = \prod_{A \subset \{1, \dots, N\}} (\sum_{k \in A} x_k)^{\theta_A} \leq N^{\sum_{A \subset \{1, \dots, N\}} \theta_A} \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

This completes the proof. $\square$

Theorem 5.5. *Let $N \geq 4$. We have*

$$\|M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{s}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{L_N(\mathbb{R}^N)}.$$

Proof. It suffices to estimate, for every chamber Γ and for every $g \in G$, the operator $M_{\chi_\Gamma \cdot f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_{\chi_{g\Gamma}}$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, y), \quad x, y \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space $L_2(\mathbb{R}^N, w(x)dx)$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space $L_2(\Gamma, w(x)dx)$.

Let $N_1 = \dim(\text{span}(R))$, $N_2 = N - N_1$ and let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. For a concrete chamber Γ , let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a linear isomorphism given in Theorem 2.2 so that $\Phi(\Gamma) = \Omega$.

It suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w(\Phi^{-1}x)dx)$. Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w_\theta(x)dx)$.

By Lemma 5.3,

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y) \right| \leq c_{\alpha, \beta, \kappa} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(B(x, |x - y|))}, \quad x, y \in \Omega.$$

The other assumptions in Theorem 4.1 follow from Lemma 5.4, Lemma 3.3 and Lemma 3.4. The assertion follows now from Theorem 4.1. $\square$

6. PROOF OF THEOREM 1.1 (i)

Lemma 6.1. *We have*

$$[T_{e_j}, M_f] = M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} O_v,$$

where

$$(\delta_v f)(x) = \frac{f(x) - (O_v f)(x)}{\langle v, x \rangle}, \quad v \in R, \quad x \in \mathbb{R}^N,$$

and where O_v is (the operation of composition with) reflection with respect to vector v .

Proof. Obvious. $\square$

Lemma 6.2. *We have*

$$\|\delta_v f\|_{L_N(\mathbb{R}^N)} \leq c_N \|f\|_{\dot{W}^{1, N}(\mathbb{R}^N)}.$$

Proof. Without loss of generality, $v = e_N$. Set

$$(D_t h)(x) = h(x_1, \dots, x_{N-1}, tx_N).$$

We have

$$\delta_v f = \int_0^1 D_t(\partial_N f) dt.$$

Thus,

$$\begin{aligned} \|\delta_v f\|_{L_N(\mathbb{R}^N)} &\leq \int_0^1 \|D_t(\partial_N f)\|_{L_N(\mathbb{R}^N)} dt = \\ &= \int_0^1 t^{-\frac{1}{N}} \|\partial_N f\|_{L_N(\mathbb{R}^N)} dt = \frac{N}{N-1} \|\partial_N f\|_{L_N(\mathbb{R}^N)}. \end{aligned}$$

$\square$

Lemma 6.3. *For every $N \geq 1$, we have*

$$\|[B, A]B^{-1}\|_{N, \infty} \leq c_N \|B^{-1}[B^2, A]B^{-1}\|_{N, \infty}.$$

Proof of Theorem 1.1 (i). We prove the assertion under *a priori* restriction $f \in C_c^\infty(\mathbb{R}^N)$. Passing to the case of $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ is standard (see e.g. [refer to our papers](#)).

We have

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] &= [T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} + T_{e_j} [\Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = \\ &= [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} - R_{\text{Dunkl},j} [\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \|[\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.3, we write

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} c_N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Note that

$$\begin{aligned} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} &= \\ &= \sum_{k=1}^N \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] R_{\text{Dunkl},k} + R_{\text{Dunkl},k} [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Combining the estimates in the last two paragraphs, we obtain

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq c'_N \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c'_N \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.1, the fact that Δ_{Dunkl} commutes with O_v and the triangle inequality, we obtain

$$\begin{aligned} \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|M_{\partial_k f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \end{aligned}$$

$$+ \frac{N}{2(N-1)} \sum_{v \in R} |v| \kappa(v) \|M_{\delta_v} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}.$$

Thus,

$$\begin{aligned} & \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq c_N'' \sum_{k=1}^N \|M_{\partial_k} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ & + c_\kappa \sum_{v \in R} \|M_{\delta_v} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

The assertion follows now from Theorem 5.5 and Lemma 6.2. $\square$

7. PROOF OF THEOREM 1.1 (ii)

Lemma 7.1. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\text{Dunkl},j}(x, y) \geq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{\text{Dunkl},j}(x, y) \leq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The proof is identical to that in [6, Theorem 1.2]. $\square$

Lemma 7.2. *Let $Q \in \mathbb{D}$. We have*

$$K_j(x, y) \geq \frac{c_k}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

Proof. The assertion follows immediately from Lemma 7.1. $\square$

Lemma 7.3. *Let real-valued $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ be such that the commutator $[R_{\text{Dunkl},j}, M_f]$ is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. For every $Q \in \mathbb{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$ and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq c_k |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Proof. Let α be the median of f on $3Q$ and β be the median of f on $Q + 3s(Q)e_j$. Assume for definiteness $\alpha \geq \beta$. Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : f(x) \geq \beta\}, \\ \xi_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{A_s}, \quad \eta_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} & \langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ & = \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If $x \in B_1$ and $y \in A_1$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If $x \in B_2$ and $y \in A_2$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot \left(w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ &= \frac{c_k}{w^2(Q)} \cdot \left(\frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ &\geq \frac{c_k}{4w(Q)} \cdot \left(\int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} &-\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ &= \frac{c_k}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} &\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy \\ &+ \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q). \end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)}|.$$

Hence,

$$c'_k |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)}|$$

and, finally,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)}|.$$

□

Lemma 7.4. *Suppose that $f \in L_1^{\text{loc}}(\mathbb{R}^N)$. If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x) dx))$ for some $j \in \{1, 2, \dots, N\}$, then*

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ & \leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x) dx))}. \end{aligned}$$

Proof. Let $\{\xi_Q\}_{Q \in \mathbb{D}}$ and $\{\eta_Q\}_{Q \in \mathbb{D}}$ be functions constructed in Lemma 7.3. By Lemma 2.6,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x) dx)} \right\}_{Q \in \mathbb{D}} \in l_p.$$

The first assertion follows now from Lemma 7.3. □

Lemma 7.5. *Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_k}{m(Q)} \int_Q w(x) dx.$$

Proof. If $x = c_Q + y$, $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$, then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

Fix chamber Γ such that $c_Q \in \Gamma$. We have

$$\int_Q w(x) dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x) dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y) dy.$$

If $y \in \Gamma$, then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone $\Gamma' \subset \Gamma$ such that (i) interior of Γ' is non-empty and (ii) $\partial\Gamma \cap \partial\Gamma' = \{0\}$. We have

$$|\langle y, v \rangle| \geq c_{\Gamma'} |y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x)dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'} |y|)^{k(v)} dy \geq \\ &\geq \int_{\frac{1}{4}s(Q) \leq |y| \leq \frac{1}{4}s(Q)} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2}c_{\Gamma'} s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4}c_{\Gamma'} s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof. $\square$

Lemma 7.6. *Suppose $1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}$. Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\left(\frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx} \right)^{\frac{1}{t}} \leq c_{t,k} \frac{m(Q)}{\int_Q w(x)dx}.$$

Proof. If $v \in R$ is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2}s(Q) \geq \frac{1}{2}|\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq 2^{(t-1)|R|k(v)} |\langle c_Q, v \rangle|^{(1-t)|R|k(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|k(v)} |\langle x_0, v \rangle|^{(1-t)|R|k(v)} \cdot s(Q)^N. \end{aligned}$$

If $v \in R$ is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let $y_0 = \langle c_Q, v \rangle$ and $z_0 = c_Q - y_0 v$. Let us introduce new variables $y = \langle x, v \rangle$ and $z = x - yv$ (it varies in the plane orthogonal to v). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq \int_{|x - c_Q| \leq \frac{\sqrt{d}}{2}s(Q)} |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq \\ &\leq \int_{|y - y_0|, |z - z_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy dz = \\ &= \left(\frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y - y_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy \leq \\ &= \left(\frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy = \end{aligned}$$

$$= s(Q)^{N+(1-t)|R|k(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|k(v)} dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq c_1 s(Q)^N (\max\{|\langle c_Q, v \rangle|, r\})^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\begin{aligned} \int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left(\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \right)^{\frac{1}{|R|}} \leq \\ &\leq c_2 s(Q)^N \prod_{v \in R} (\max\{|\langle c_Q, v \rangle|, s(Q)\})^{(1-t)k(v)}. \end{aligned}$$

The assertion follows easily. $\square$

Lemma 7.7. *Measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [9].*

Proof. That both measures are doubling is obvious.

Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). By Lemmas 7.5 and 7.6,

$$\begin{aligned} \left(\frac{1}{m(Q)} \int_Q w^t(x) dx \right)^{\frac{1}{t}} &\leq \frac{c_k}{m(Q)} \int_Q w(x) dx, \quad t > 0. \\ \left(\frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}. \end{aligned}$$

The assertion follows now from Proposition 3.6 in [9]. $\square$

Proof of Theorem 1.1 (ii). The left hand side in Lemma 7.4 is the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. So, the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

Lemma 7.7 asserts that measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [9]. Appealing to Proposition 1.2 in [9], we conclude that the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the one with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. By the preceding paragraph, the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

That oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is equivalent to the Sobolev semi-norm of the function f is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [10]. The assertion follows now from the preceding paragraph. $\square$

8. PROOF OF SMOOTHNESS

Theorem 8.1. *Let $f \in C_c^\infty(\mathbb{R}^N)$.*

(i) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 = n$, the operator¹*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$;

(ii) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 > n$, the operator²*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$;

Lemma 8.2. *We have*

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N M_{f_j^{(1)}} T_{e_j} + \sum_{j=1}^N \sum_{v \in R} c_{j,v} M_{f_v^{(1)}} (T_{e_j} O_v + O_v T_{e_j}) + \\ &\quad + M_{f^{(2)}} + \sum_{v \in R} M_{f_v^{(2)}} O_v + \sum_{v, v' \in R} M_{f_{v,v'}} O_{v'} O_v. \end{aligned}$$

All the functions on the right hand side are smooth and compactly supported.

Proof. We have

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f] + [T_{e_j}, M_f] T_{e_j} = \\ &= \sum_{j=1}^N T_{e_j} M_{\partial_j f} + M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (T_{e_j} M_{\delta_v f} O_v + M_{\delta_v f} O_v T_{e_j}) = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} O_v + M_{\delta_v f} O_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) [T_{e_j}, M_{\delta_v f}] O_v = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} O_v + M_{\delta_v f} O_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N M_{\partial_j^2 f} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} \alpha_j \kappa(v) M_{\delta_v(\partial_j f)} O_v + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\partial_j(\delta_v f)} O_v + \frac{1}{4} \sum_{j=1}^N \sum_{v, v' \in R} v_j v'_j \kappa(v) \kappa(v') M_{\delta_{v'} \delta_v f} O_{v'} O_v. \end{aligned}$$

□

¹Commutators are repeated n times

²Commutators are repeated n times

Lemma 8.3. *For every $f \in C_c^\infty(\mathbb{R}^N)$, the operator*

$$[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

is bounded and

$$(1 + \Delta_{\text{Dunkl}})^{-1}[\Delta_{\text{Dunkl}}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove only the first assertion. The proof of the second one is similar (modulo Theorem 5.5).

Since each T_{e_j} commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$, each O_v commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$ and since the operators

$$\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, \quad 1 \leq j \leq N,$$

are bounded, the assertion follows from Lemma 8.2. $\square$

Lemma 8.4. *For every $n \in \mathbb{N}$, the operator³*

$$[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}$$

is bounded.

Proof. Denote for brevity

$$\text{Expr}_n(f) = [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}.$$

We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, the operator $\text{Expr}_n(f)$ is bounded." Base of induction is given by Lemma 8.3. It suffices to prove the step of induction.

Since each T_{e_j} and each O_v commutes with Δ_{Dunkl} , it follows from Lemma 8.2 that

$$\begin{aligned} \text{Expr}_{n+1}(f) &= \\ &= \sum_{j=1}^N \text{Expr}_n(f_j^{(1)}) \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ \sum_{j=1}^N \sum_{v \in R} c_{j,v} \text{Expr}_n(f_v^{(1)}) \cdot \left(\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} O_v + O_v \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \right) + \\ &+ \text{Expr}_n(f^{(2)}) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \sum_{v \in R} \text{Expr}_n(f_v^{(2)}) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_v + \\ &+ \sum_{v,v' \in R} \text{Expr}_n(f_{v,v'}^{(2)}) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_{v'} O_v. \end{aligned}$$

In each summand, the first factor is bounded by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Proof of Theorem 8.1 (i). The assertion follows from Lemma 8.4 and Hadamard 3 lines theorem. $\square$

Lemma 8.5. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-1} \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

³Commutators are repeated n times

Proof. Set

$$\mathcal{D} = \sum_{j=1}^N \gamma_j \otimes T_{e_j},$$

where $\{\gamma_j\}_{j=1}^N$ are Pauli matrices. We have

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= (1 \otimes M_f)(\mathcal{D}^2 + 1)^{-1} = (1 \otimes M_f)(\mathcal{D} + i)^{-1}(\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + [1 \otimes M_f, (\mathcal{D} + i)^{-1}](\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}[\mathcal{D}, 1 \otimes M_f](\mathcal{D}^2 + 1)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}\left(\sum_{j=1}^N \gamma_j \otimes [T_{e_j}, M_f]\right)(\mathcal{D}^2 + 1)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(\gamma_j \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} + \\ &\quad + \sum_{j=1}^N \sum_{\alpha \in R_+} \alpha_j \kappa(\alpha)(\mathcal{D} + i)^{-1}(\gamma_j \otimes M_{f_\alpha} O_\alpha)(\mathcal{D}^2 + 1)^{-1}. \end{aligned}$$

Since Δ_{Dunkl} commutes with O_α , it follows that

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes 1) + \\ &\quad + \sum_{j=1}^N \sum_{\alpha \in R_+} \alpha_j \kappa(\alpha)(\mathcal{D} + i)^{-1}(1 \otimes M_{f_\alpha})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes O_\alpha). \end{aligned}$$

Clearly,

$$\begin{aligned} &\|(\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} = \\ &= \|(\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}} M_f (\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &= \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq 2^{\frac{2}{N}} \|M_{|f|^{\frac{1}{2}}} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))}^2 \leq \\ &\stackrel{\text{Th. 5.5}}{\leq} c_{N, \kappa} \| |f|^{\frac{1}{2}} \|_{L_N(\mathbb{R}^N)}^2 = c_{N, \kappa} \|f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)} \end{aligned}$$

and, similarly,

$$\begin{aligned} &\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\partial_j f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}, \\ &\|(\mathcal{D} + i)^{-1}(1 \otimes M_{f_\alpha})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|f_\alpha\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}. \end{aligned}$$

This completes the proof. $\square$

Lemma 8.6. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))."$$

Base of induction (i.e. cases $n = 1$ and $n = 2$) are given by Theorem 5.5 and Lemma 8.5. It suffices to prove the step of induction.

We have

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + [M_f, (1 + \Delta_{\text{Dunkl}})^{-1}](1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}}. \end{aligned}$$

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

Thus,

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = (1 + \Delta_{\text{Dunkl}})^{-1} M_f \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + \\ &+ (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

In the first summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.5 and the second one belongs to the ideal $\mathcal{L}_{\frac{N}{n-1}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption. In the second summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.3, the second factor belongs to the ideal $\mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the third one is obviously bounded. $\square$

Lemma 8.7. *Let $k, l \in \mathbb{Z}_+$ with $l > k$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. We prove the assertion by induction on k . Base of induction (i.e., the case $k = 0$) is established in Lemma 8.6.

We have

$$\begin{aligned} & T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f T_{e_{m_k}}(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} [T_{e_{m_k}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = \\ &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times \frac{T_{e_{m_k}}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\partial_{m_k} f}(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ &+ \frac{1}{2} \sum_{v \in R} v_{m_k} \kappa(v) T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\delta_v f}(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_v. \end{aligned}$$

In each summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Lemma 8.8. *Let $s_1, s_2 \in \mathbb{R}$ be such that $s_1 + s_2 > 0$. We have*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. Step 1: Let us prove the assertion for $s_1, s_2 \in \mathbb{Z}$.

Suppose first $s_1 \leq 0$. Denote $k = -s_1 \geq 0$ and $l = s_2 > k$. Denote for brevity

$$V = (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} (1 + \sum_{j_1, \dots, j_k=1}^N |T_{e_{j_1}} \cdots T_{e_{j_k}}|)^{-1}.$$

Obviously, V is bounded. We have

$$\begin{aligned} (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} &= V \times (1 + \sum_{m_1, \dots, m_k=1}^N |T_{e_{m_1}} \cdots T_{e_{m_k}}|) = V + \\ &+ \sum_{m_1, \dots, m_k=1}^N V \times \text{sgn}(T_{e_{m_1}} \cdots T_{e_{m_k}})^* \times T_{e_{m_1}} \cdots T_{e_{m_k}}. \end{aligned}$$

It, therefore, suffices to prove that operators

$$M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}, \quad T_{e_{m_1}} \cdots T_{e_{m_k}} M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This is established in Lemma 8.6 and Lemma 8.7, respectively. Hence, the claim is proved for $s_1, s_2 \in \mathbb{Z}$ and $s_1 \leq 0$.

If $s_2 \leq 0$, then the claim follows from that for $s_2 \leq 0$ by taking the adjoint. If $s_1, s_2 \geq 0$, then, according to Lemma 8.7 the operators

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}} M_f, \quad M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. The claim for $s_1, s_2 \geq 0$ follows now from the Hadamard 3 lines theorem.

Step 2: The general case follows now from Step 1 by complex interpolation. $\square$

Proof of Theorem 8.1 (ii). Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

We have

$$\begin{aligned} &(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-s_1}{2}} \times \\ &\quad \times (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1-n}{2}} M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}. \end{aligned}$$

The first factor is bounded by Theorem 8.1 (i). The second factor belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.8. $\square$

9. FIRST APPROXIMATION THEOREM

Theorem 9.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{1}{2}[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in \\ & \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Lemma 9.2. *Let $B \geq 1$. Suppose*

$$B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

We have

$$[B, A]B^{-1} - \frac{1}{2}[B^2, A]B^{-2} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B = \frac{2}{\pi} \int_0^\infty \frac{B^2 d\lambda}{\lambda^2 + B^2}.$$

Hence,

$$[B, A]B^{-1} = \frac{2}{\pi} \int_0^\infty \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} d\lambda.$$

Obviously,

$$\begin{aligned} \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} &= - \left[\frac{\lambda^2}{\lambda^2 + B^2}, A \right] B^{-1} = \frac{\lambda^2}{\lambda^2 + B^2} [B^2, A] \frac{1}{B(\lambda^2 + B^2)} = \\ &= [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} - \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2}. \end{aligned}$$

Obviously,

$$\begin{aligned} \int_0^\infty [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} d\lambda &= [B^2, A] B^{-1} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + B^2)^2} d\lambda = \\ &= [B^2, A] B^{-2} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + 1)^2} d\lambda = \frac{\pi}{4} [B^2, A] B^{-2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty} \left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty. \end{aligned}$$

Next,

$$\left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{\lambda^2 t}{\lambda^2 + t^2} \leq \lambda$$

and

$$\left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{t^{\frac{3}{2}}}{(\lambda^2 + t^2)^2} \leq (\lambda^2 + 1)^{-\frac{5}{4}}.$$

Thus,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \frac{\lambda}{(\lambda^2 + 1)^{\frac{5}{4}}} \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty}. \end{aligned}$$

Hence, the integral

$$\int_0^\infty \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} d\lambda$$

converges absolutely in $\mathcal{L}_{\frac{2N}{3}, \infty}$ and the assertion follows. $\square$

Proof of Theorem 9.1. The assertion follows from Theorem 8.1 and Lemma 9.2. $\square$

10. SECOND APPROXIMATION THEOREM

Theorem 10.1. *If $f \in C_c^\infty(\mathbb{R}^N)$, then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Lemma 10.2. *Let A, B_1, B_2 be such that $B_1, B_2 \geq 1$. Suppose that*

- (i) $A(B_2 - B_1)B_2^{-\frac{1}{2}}, [B_1, A]B_1^{-\frac{1}{2}}$ and $[B_1, A]B_2^{-\frac{1}{2}}$ are bounded;
- (ii) *there exists A' with $B_1^{-\frac{3}{4}}A' \in \mathcal{L}_{\frac{2N}{3}, \infty}$ such that $A = A'A$ and $[B_1, A] = A'[B_1, A]$;*

We have

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B_k^{-\frac{1}{2}} = \frac{2}{\pi} \int_0^\infty \frac{d\lambda}{\lambda^2 + B_k}, \quad k = 1, 2.$$

Thus,

$$AB_1^{-\frac{1}{2}} - AB_2^{-\frac{1}{2}} = \int_0^\infty A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) d\lambda.$$

Obviously,

$$\begin{aligned} A \left(\frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_2} \right) &= A \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A (B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{1}{\lambda^2 + B_1} A (B_2 - B_1) \frac{1}{\lambda^2 + B_2} + \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_1} - \frac{1}{\lambda^2 + B_1} [B_1, A] \frac{1}{\lambda^2 + B_2}. \end{aligned}$$

Consider, for example, the first summand. We write

$$\begin{aligned} \frac{1}{\lambda^2 + B_1} A (B_2 - B_1) \frac{1}{\lambda^2 + B_2} &= \frac{1}{\lambda^2 + B_1} A' \cdot A (B_2 - B_1) \frac{1}{\lambda^2 + B_2} = \\ &= \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \cdot B_1^{-\frac{3}{4}} A' \cdot A (B_2 - B_1) B_2^{-\frac{1}{2}} \cdot \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2}. \end{aligned}$$

Thus,

$$\begin{aligned} &\left\| \frac{1}{\lambda^2 + B_1} A (B_2 - B_1) \frac{1}{\lambda^2 + B_2} \right\|_{\frac{3N}{4}, \infty} \leq \\ &\leq \left\| \frac{B_1^{\frac{3}{4}}}{\lambda^2 + B_1} \right\|_\infty \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1)B_2^{-\frac{1}{2}}\|_\infty \cdot \left\| \frac{B_2^{\frac{1}{2}}}{\lambda^2 + B_2} \right\|_\infty \leq \\ &\leq \sup_{t \geq 1} \frac{t^{\frac{3}{4}}}{\lambda^2 + t} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1)B_2^{-\frac{1}{2}}\|_\infty \cdot \sup_{t \geq 1} \frac{t^{\frac{1}{2}}}{\lambda^2 + t} \leq \\ &\leq (\lambda^2 + 1)^{-\frac{1}{4}} \cdot \left\| B_1^{-\frac{3}{4}} A' \right\|_{\frac{3N}{4}, \infty} \cdot \|A(B_2 - B_1)B_2^{-\frac{1}{2}}\|_\infty \cdot (\lambda^2 + 1)^{-\frac{1}{2}}. \end{aligned}$$

Hence, the first summand is absolutely integrable in $\mathcal{L}_{\frac{3N}{4}, \infty}$. Similarly, so are the other summands. This completes the proof. $\square$

Lemma 10.3. *Let P be a differential operator of degree 1 with smooth compactly supported coefficients. Suppose that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. The operators*

$$P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}, \quad P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$.

Proof of Theorem 10.1. Step 1: Suppose first that f is supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$.

We apply Lemma 10.2 with $A = M_f$, $B_1 = 1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}$ and $B_2 = 1 + \Delta_{\text{Dunkl}}$.

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $f = f\phi$ and set $A' = M_\phi$. We have

$$M_\phi[M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f].$$

Thus, $A = A'A$ and $[B_1, A] = A'[B_1, A]$. It follows from the usual Cwikel estimates that $B_1^{-\frac{3}{4}} A' \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This verifies the assumption (ii) in Lemma 10.2.

Note that $[B_1, A]$ is a differential operator P of degree 1 with smooth compactly supported coefficients. Furthermore, the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$. By Lemma 10.3, the operators

$$[B_1, A]B_1^{-\frac{1}{2}} = P(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \quad [B_1, A]B_2^{-\frac{1}{2}} = P(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}},$$

are bounded on $L_2(\mathbb{R}^N, w(x)dx)$. Note that there exist differential operators $\{P_g\}_{g \in G}$ of degree 1 with smooth compactly supported coefficients such that the coefficients are supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$ and such that

$$A(B_2 - B_1) = \sum_{g \in G} P_g U_g.$$

Since $\{U_g\}_{g \in G}$ commute with Δ_{Dunkl} , it follows that

$$A(B_2 - B_1) = \sum_{g \in G} P_g (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \cdot U_g$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. This verifies the assumption (i) in Lemma 10.2.

Using Lemma 10.2, we infer the claim of Step 1.

Step 2: Choose a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ such that (a) $0 \leq \phi_n \leq 1$ for $n \geq 1$ (b) ϕ_n is supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, for $n \geq 1$ (c) $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$.

By Step 1,

$$M_{f\phi_n} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n} (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

for every $n \geq 1$.

By Theorem 5.5,

$$M_{f\phi_n} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \rightarrow M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. By the usual Cwikel estimate,

$$M_{f\phi_n} (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \rightarrow M_f (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. Thus

$$\begin{aligned} & \lim_{n \rightarrow \infty} M_{f\phi_n} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n} (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = \\ & = M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \end{aligned}$$

where the limit is understood in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since the separable part $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

11. FINAL APPROXIMATION THEOREM

Theorem 11.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & M_{w^{\frac{1}{2}}} [R_{\text{Dunkl},j}, M_f] M_{w^{-\frac{1}{2}}} \in \\ & \in M_{\partial_j f} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_v - \\ & \quad - \frac{\partial_j}{(1 + \Delta)^{\frac{1}{2}}} \cdot \sum_{k=1}^N M_{\partial_k f} \frac{\partial_k}{1 + \Delta} - \\ & \quad - \frac{1}{4} \frac{\partial_j}{1 + \Delta} \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{\partial_k}{1 + \Delta} O_v + O_v \frac{\partial_k}{1 + \Delta} \right) + \\ & \quad + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N)). \end{aligned}$$

Lemma 11.2. *We have*

$$[R_{\text{Dunkl},j}, M_f] - \left[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f \right] \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$h(x) = 1 - \frac{|x|}{(1 + |x|^2)^{\frac{1}{2}}}, \quad x \in \mathbb{R}^N.$$

We, obviously, have

$$R_{\text{Dunkl},j} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}).$$

Thus,

$$\begin{aligned} & [R_{\text{Dunkl},j}, M_f] - \left[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f \right] = \\ & = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) M_f - M_f h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j} = \\ & = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) (1 + \Delta_{\text{Dunkl}}) \cdot (1 + \Delta_{\text{Dunkl}})^{-1} M_f - \\ & \quad - M_f (1 + \Delta_{\text{Dunkl}})^{-1} \cdot (1 + \Delta_{\text{Dunkl}}) h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j}. \end{aligned}$$

In the first summand on the right hand side, the first and the second factors are bounded and the third one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. In the second summand on the right hand side, the second and the third factors are bounded and the first one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. $\square$

Lemma 11.3. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} - 2 \sum_{j=1}^N M_{\partial_j f} \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} - \\ & - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} \left(\frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} O_v + O_v \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \right) \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Proof. The assertion follows from Lemma 8.2 and Lemma 8.6. $\square$

Lemma 11.4. *For every $f \in C_c^\infty(\mathbb{R}^N)$ supported away from the hyperplanes $\{\{x : \langle x, v \rangle = 0\}\}_{v \in R}$, we have*

$$M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} - M_f \frac{M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Obviously,

$$M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} = M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \times \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}.$$

By Theorem 10.1,

$$\begin{aligned} & M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\ & \in M_f (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \times \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Since

$$[M_f, (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}] \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)),$$

it follows that

$$\begin{aligned} & M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\ & \in (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \times M_f T_{e_j} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Clearly,

$$M_f T_{e_j} - M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}} \cdot M_f = \sum_{g \in G} M_{f_g} U_g,$$

where functions $\{f_g\}_{g \in G}$ belong to $C_c^\infty(\mathbb{R}^N)$. Since U_g commutes with Δ_{Dunkl} and since

$$(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \times M_{f_g} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)),$$

it follows that

$$\begin{aligned} & M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\ & \in (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \times M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}} \times M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ & + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Again applying Theorem 10.1, we conclude that

$$\begin{aligned} & M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} \in \\ & \in (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \times M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}} \times M_f (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} + \\ & + (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

The assertion follows now from the usual commutator inclusions. $\square$

Lemma 11.5. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f \frac{T_{e_j}}{1 + \Delta_{\text{Dunkl}}} - M_f \frac{M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Using Lemma 11.4 and repeating the argument in the proof of Theorem 10.1 *mutatis mutandi*, we complete the proof. $\square$

Lemma 11.6. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} - 2 \sum_{j=1}^N M_{\partial_j f} \frac{M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} - \\ & - \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} \left(\frac{M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} O_v + O_v \frac{M_{w^{-\frac{1}{2}}} \partial_j M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right) \in \\ & \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Proof. The assertion follows immediately from Lemma 11.5 and Lemma 11.3. $\square$

Proof of Theorem 11.1. We have

$$\begin{aligned} & \left[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f \right] = \\ & = [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

Using Lemma 11.2 and Theorem 9.1, we write

$$\begin{aligned} & [R_{\text{Dunkl},j}, M_f] \in [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \\ & - \frac{1}{2} \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Using Lemma 8.6 and Lemma 6.1 for the first summand and Lemma 11.6 for the second summand, we obtain

$$\begin{aligned} & [R_{\text{Dunkl},j}, M_f] \in M_{\partial_j f} (1 + M_{w^{-\frac{1}{2}}} \Delta_{\text{Dunkl}} M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} + \\ & + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} (1 + M_{w^{-\frac{1}{2}}} \Delta_{\text{Dunkl}} M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} O_v - \\ & - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N M_{\partial_k f} \frac{M_{w^{-\frac{1}{2}}} \partial_k M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} - \\ & - \frac{1}{4} \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot \sum_{k=1}^N \sum_{v \in R} v_k \kappa(v) M_{\delta_v f} \left(\frac{M_{w^{-\frac{1}{2}}} \partial_k M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} O_v + O_v \frac{M_{w^{-\frac{1}{2}}} \partial_k M_{w^{\frac{1}{2}}}}{1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right) \in \\ & \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Repeating the argument in Lemmas 11.4, 11.5 and 11.6, we complete the proof. $\square$

12. SPECTRAL ASYMPTOTIC FORMULA

APPENDIX A. ZHIJIE LEMMA

Lemma A.1. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. Then there exists a constant $C > 0$ such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s, \infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in \mathcal{E}} Q$ we may choose a cube $Q_x \in \mathcal{E}$ of minimal volume such that $x \in Q_x$. Every $Q \in \mathcal{E}$ containing x is then a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix $t > 0$ and let $\mathcal{E}$ be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since $\mathcal{E}$ is arbitrary,

$$\#\left\{ Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t \right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak ℓ^s estimate follows. This completes the proof of Lemma A.1. $\square$

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ZHIJIE FAN, SCHOOL OF MATHEMATICAL SCIENCES, SOUTH CHINA NORMAL UNIVERSITY, GUANGZHOU 510631, CHINA

Email address: fanzhj3@mail2.sysu.edu.cn

JI LI, DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW2109, AUSTRALIA

Email address: ji.li@mq.edu.au

SCHOOL OF MATHEMATICS AND STATISTICS, HNP-LAMA, CENTRAL SOUTH UNIVERSITY, CHANGSHA 410075, CHINA

Email address: d.zanin@csu.edu.cn